and fecal matter. The most common group of indicator organisms used in water quality monitoring are the coliforms, bacteria that are Gram negative, aerobic or facultative anaerobic, non-spore forming rods that ferment lactose with gas production within 48 hours at 35°C. Examples of coliforms are *Escherichia coli, Enterobacter aerogenes* and *Klebsiella pneumonia*. Two analytical procedures were followed to check the presence of coliforms in water. They are Most Probable Number (MPN) and Membrane Filteration (MF) technique. The number of coliforms per 100ml of water sample is estimated to find the quality of water and its suitability for drinking purposes. In addition to coliforms, coli phages, *Clostridia* and human enteric viruses are also monitored in drinking water.

Waterborne diseases

Waterborne diseases are posing a serious threat to health (Table 9.3).

Table 9.3: Waterborne diseases

Waterborne diseases	Causative agent	Symptoms
Bacterial diseases		
Enteric fever	*Salmonella typhi*	Fever & enlargement of spleen
Cholera	*Vibrio cholerae*	Vomiting & watery diarrhea
Leptospirosis (Weils Disease)	*Leptospira interrogans*	High fever, red eyes, muscle pain and vomiting
Viral diseases		
Infectious Hepatitis	*Hepatitis A*	Jaundice, vomiting & abdominal pain
Gastroenteritis	*Rotavirus*	Diarrhoea, vomiting
Poliomyelitis	*Coxsackie Virus*	Head ache, neck stiffness, flaccid paralysis.
Protozoan diseases		
Giardiasis	*Giardia lamblia*	Chronic diarrhoea, abdominal cramp, fatigue & weight loss
Amoebiasis	*Entamoeba histolytica*	Stomach pain, bloody stools, fever
Meningoencephalitis	*Naegleria fowleri*	Ulceration, watery, bloody diarrhoea
Treamatode disease		
Schistosomiasis	*Schistosoma*	Drowsy, confusion, head ache, stiff neck, Diarrhoea

9.5 Sewage Treatment

Wastewater treatments also called **sewage treatment** which removes the impurities from wastewater, or sewage, before disposal into natural bodies of water. In broad terms, water is said to be polluted when it contains enough impurities to make it unfit for a particular use, such as

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